

House Price Prediction Approach based on Deep Learning and ARIMA Model

Feng Wang, Yang Zou, Haoyu Zhang and Haodong Shi
College of Information Science and Engineering
Henan University of Technology
 Zhengzhou, China
 wangfeng_haut@haut.edu.cn

Abstract—The nonlinear relationship between influential factors and house price and inadequate number of sample size could be the cause of the poor performance of the traditional models. Meanwhile, the daily data of the real estate market is very huge and it is increasing rapidly. The traditional house price prediction approaches are lack of capacity for massive data analysis, causing low utilization of data. To address these concerns, a house price prediction model based on deep learning is proposed in this paper, implemented on the TensorFlow framework. Adam optimizer is used to train the model, where Relu function is adopted to be the activation function. Then house price trend is predicted based on the ARIMA model. By using Scrapy, housing data are obtained from Internet to be the experimental dataset. Comparative experiments were conducted between the proposed approach and SVR method. The experimental results show that individual house price predicted by the proposed approach is better than that of SVR method. And the predicted house price trend is mainly agreement with the real situation.

Keywords—house price prediction, deep learning, ARIMA model

I. INTRODUCTION

Housing price is an important reflection of the economy and an important indicator for healthy and stable development of the real estate. Meanwhile, housing price is also one of the hot issues that the whole society focuses on because it will affect the interests of all quarters. Effective prediction of house price plays a significant role in shaping the economy. First, it is helpful for government to better regulate the real estate market and maintain its stable, healthy and orderly development. Second, it can help real estate developers to make investment decisions ahead of time. Third, housing is a shelter to fulfill the fundamental need of an individual, and it can also be a form of investment. Obviously, house price ranges are of great interest for both buyers and sellers. The forecast of price is important for them to make informed decisions. With realistic price estimates, they could acquire information of the house price trends before executing any transactions.

Traditionally, prediction of house price is often determined by professional appraisers. But an appraiser is likely to be biased due to vested interest from the lender, mortgage broker, buyer, or seller. Therefore, an automated prediction system is helpful to serve as an independent third party source that may be less biased. And there is evidence

that house price is forecastable to a certain degree [1]. House price is a form of time series. Various techniques have been proposed in predicting house price. A house price prediction model tries to understand the influential factors controlling the changes of price over a given area. Several models based on traditional statistics approach were proposed in [2]. Hedonic regression and geographically weighted regression are two common approaches in dealing with spatial data and with house price [3]. Obviously, the driving forces for house price are complex and inter-related processes that traditional statistical approaches fail to address. Although the hedonic price model has been widely recognized over the past years, it also received a lot of criticisms on model assumption and estimation as well as on solving nonlinear problem, global regression and local clustering [4, 5]. Therefore, nonlinear machine learning and fuzzy logics were used to predict the variation of house price [6]. Neural Network was employed for property prediction in [2, 7]. Support Vector Machine was used in combination with optimization models such as Generic Algorithm and Particle Swarm Optimization [8, 9]. Different machine learning algorithms such as RIPPER (Repeated Incremental Pruning to Produce Error Reduction), Naïve Bayes and AdaBoost were analyzed in [10]. It concluded that RIPPER algorithm outperforms other models on predicting house price. Linear regression, Decision Tree and Nearest Neighbor were applied to predict the house price in [11]. And the study showed that Naïve Bayes was consistent for unequal distribution frequency and Decision Tree was the most consistent classifier for equal frequency distributions. The error in linear regression was little high, but it predicted numerical values of selling price instead of a range of selling price as some other classifiers do.

In addition, machine vision methods are also adopted to extract visual features in estimating the house price model. Traditional housing features were supplemented with visual features extracted from property photos to predict house price in [12]. Reference [13] employed a novel recurrent neural network LSTM model to predict house price using both the location on a random walk sequence and the visual features in an end-to-end learning model. The prediction result was improved by making use of traditional housing features as well as visual features from Google Street View images and Bing aerial images that can capture the urban quality of a neighborhood [14].

House price prediction is a challenging problem. On the one hand, the influential factors of house price are complex and changed nonlinearly, making the traditional models usually have big prediction errors. On the other hand, the daily data of the real estate market is very huge and it is increasing rapidly. The traditional prediction approaches are lack of capacity for massive data analysis, causing low

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utilization of data. To address these concerns, a house price prediction model based on deep learning is proposed first. Then house price trend is predicted by using the ARIMA model. To demonstrate the effectiveness and utility of the proposed approach, comparative experiments were conducted on the real housing data extracted from Internet.

II. DESIGN AND IMPLEMENTATION

A. Feature Selection and Data Preprocessing

There are many factors that affect the house price and they are complex and volatile. In order to reflect the changing rules of house price objectively, 13 explanatory features are selected to be the influential factors of house price. These features include size, location, orientation, numbers of bedrooms, numbers of rooms, numbers of bathrooms, types of the floors, types of the decoration, accessibility to schools, availability of elevator, built year, accessibility to metro and metro distance.

The raw housing data are firstly extracted from Internet. Then a data preprocessing step is performed to transform the raw data to produce outputs that can be smoothly used as inputs in data modeling. During data preprocessing, outliers were firstly determined by inspection of the distribution of values, and subsequently removed/corrected. In the raw dataset, the formats and attributes of the features are different. There are numerical data and categorical features. The missing numerical variables are estimated by the mean of the corresponding features. For missing values in categorical variables, they are filled with the default ones. Next data discretization is performed. In order to make the data with different format usable for the prediction model, categorical data was converted into separated indicator data. In addition, the data may have different units and orders of magnitude. In order to eliminate the influence of external factors on the prediction results, normalization operation is then performed to unify the numerical variables.

B. Prediction of Individual House Price

A deep learning model is used to predict the price of an individual house. The model consists of one input data layer (example the property attributes), four hidden layers and one output layer (estimated house price). The samples with 13 housing features are fed into the input layer and then transferred through the hidden layers to the output layer. Each computational unit has a set of input connections that receive signals from other computational units and a bias adjustment, a set of weights for input connection and bias adjustment, and a nonlinear function known as an activation function or transfer function that transforms the sum of the weighted inputs and bias to decide the value of the output from computational unit. For the computation unit (node j), its output is the result of applying an activation function to the summation of all signals from each connection (a_i) times the value of the connection weight between node j and connection i (w_{ji}), as shown in (1) - (3):

$$S_j = \sum_i w_{ji} a_i, \quad (1)$$

$$O_j = R(S_j), \quad (2)$$

where O_j is output for node j and R is activation function that takes the form of ReLU function in the proposed model, i.e.

$$\text{ReLU}(x) = \begin{cases} x, & \text{if } (x > 0) \\ 0, & \text{otherwise} \end{cases}. \quad (3)$$

Before applying the model to predict an individual house price, it has to be trained first. The training process aims to modify the parameters of the model to reduce prediction errors by comparison between the predicted house price and the actual price. The propagation process of data in the network consists of forward propagation and back propagation. Forward propagation is firstly used to produce an output (estimated house price) for a particular input after the nonlinear process of four hidden layers. Then, the model compares the model output with the actual output (actual house price). The accuracy of this value is determined by the total mean square error (MSE) which is treated as the loss function or cost function for the model, as shown in (4):

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (P_i - P'_i)^2, \quad (4)$$

where P_i is actual house price, P'_i is predicted house price and n is the number of observations. Back propagation is then applied in an attempt to reduce prediction errors, which is done through the adjusting of the connection weights. Adam optimizer is employed to train the network. And the relationship between influential factors and house price is learned by iteratively updating the parameters in hidden layers (i.e., input weights and biases).

The performance of the network can be influenced by the number of hidden layers and nodes included in each hidden layer. Unfortunately, there exists little theory to determine the optimal number of hidden layers and nodes, and also the optimal internal error threshold. Therefore, a trial-and-error process is applied to find the optimal model.

C. Prediction of House Price Trend

ARIMA (Autoregressive Integrated Moving Average) model is a well-known time series forecasting method. It is assumed that past values of the series plus previous error terms contain information for the purpose of forecasting. The main advantages of ARIMA model include it requires data on the time series in question only and it has good short-run forecasting ability. Therefore, ARIMA model is employed in this paper to predict the trend of house price.

ARIMA model is the combination of the AR (autoregressive model), MA (moving average model) and ARMA models, where AR uses previous data to predict future change, MA tries to reduce the forecasting errors, AR and MA are mixed to be the ARMA model. P order AR and Q order MA can be denoted by (5) and (6) respectively:

$$u_t = c + \sum_{i=1}^p \phi_i u_{t-i} + \varepsilon_t, \quad (5)$$

$$u_t = \mu + \sum_{i=1}^q \theta_i \varepsilon_{t-i} + \varepsilon_t, \quad (6)$$

where c and μ are two constants, p is the order of AR model, ϕ_i is AR coefficient, ε_i is white noise series with a mean value of zero and a variance of δ^2 , q is the order of MA model, θ_i is MA coefficient. Based on AR model and MA model, ARMA model can be denoted by (7):

$$u_t = c + \sum_{i=1}^p \phi_i u_{t-i} + \varepsilon_t + \sum_{i=1}^q \theta_i \varepsilon_{t-i} . \quad (7)$$

The ARMA model can be used when the time series is stationary, i.e. a more homogeneous mean and variance. But most original time series can't satisfy this condition. The ARIMA model hasn't that limitation. For nonstationary time series, an initial differencing step can be used to make the obtained series stationary so that the ARMA model can then be applied. If y_t series after d times differencing becomes a stationary one and it is not stationary in $d - 1$ times differencing, the y_t series is called d order integration series. And y_t satisfy the modeling condition of ARIMA(p, d, q) model, where d is the degree of differencing.

III. EXPERIMENTAL RESULTS

The proposed approach was implemented on the TensorFlow framework. By using Scrapy, the experimental dataset was obtained from the website of Lianjia, which is one of Chinese leading property agents. A total of 11,937 samples were extracted from the website, where 10,937 of them were used to be the training data and the rest 1,000 to be the test data.

A. Prediction of Individual House Price

During the training stage, the learning rate is set to be 0.01 and the keep_prob hyperparameter for dropout is set to be 0.9 in order to reduce overfitting.

Fig. 1 shows the change curves of training loss and test loss, where the bold curve and thin curve denote test loss and training loss respectively. From Fig. 1, it can be found that both the training loss and test loss decrease with the increase of iteration and their change curves are similar.

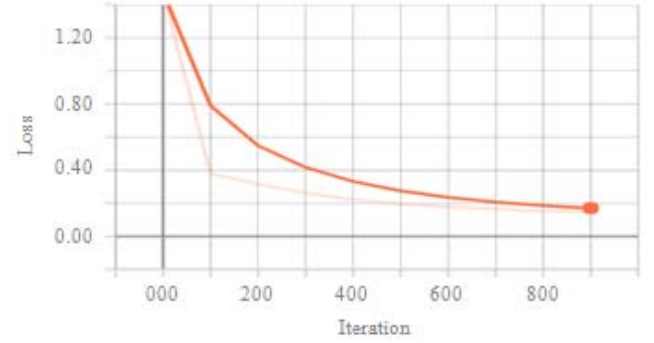


Fig. 1. The change curve of training loss and test loss.

To evaluate the performance of the proposed approach, a support vector regression (SVR) model is adopted to perform the comparison experiments. And the experimental results are shown in Figs. 2 - 5, where the solid and dashed curves denote the ground truth and predicted values respectively. For convenience, only the first 100 samples are illustrated.

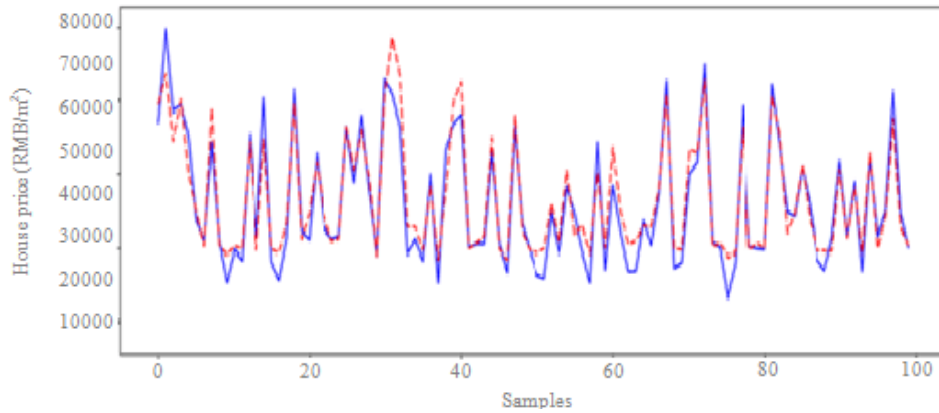


Fig. 2. The ground truth and predicted value of the proposed model on training dataset.

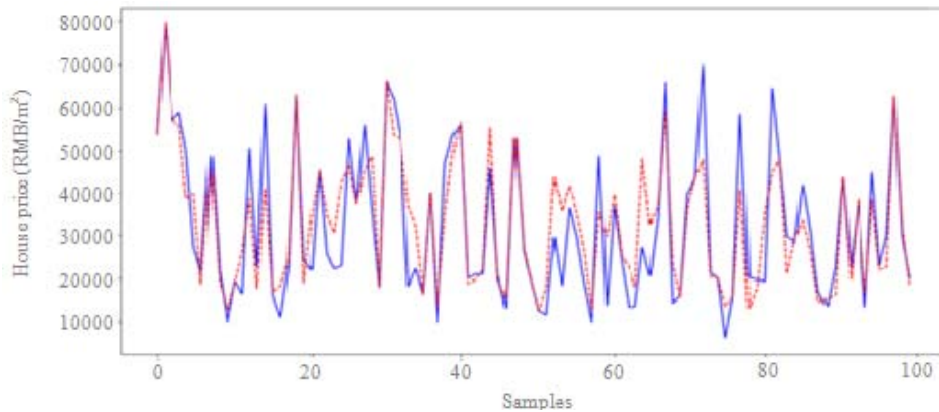


Fig. 3. The ground truth and predicted value of the SVR model on training dataset .

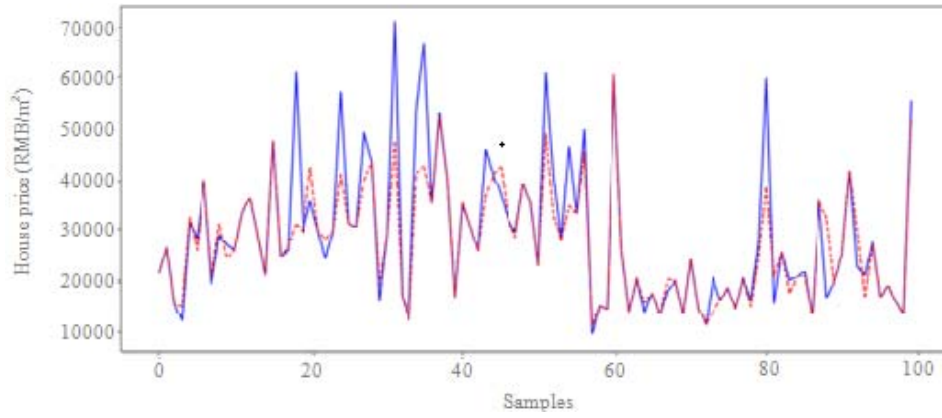


Fig. 4. The ground truth and predicted value of the proposed model on test dataset.

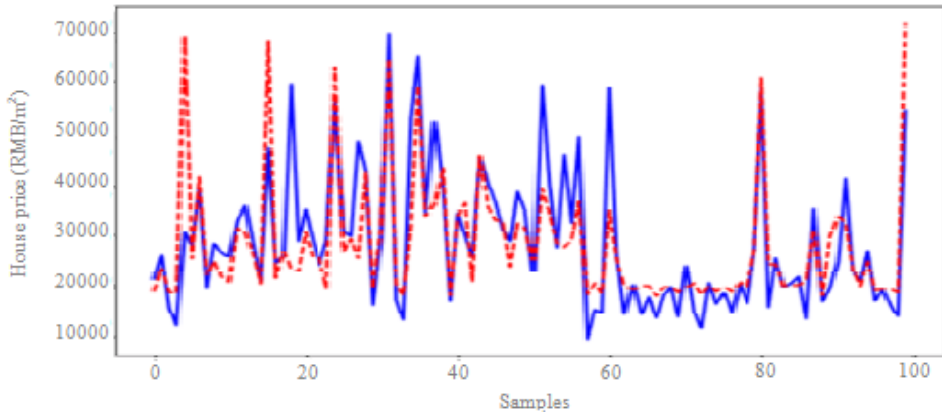


Fig. 5. The ground truth and predicted value of the SVR model on test dataset.

Figs. 2 and 4 show comparisons between the proposed model fit data and ground truth on training dataset and test dataset respectively. Correspondingly, Figs. 3 and 5 show comparisons between the SVR model fit data and ground truth on training dataset and test dataset respectively.

The root mean square error (RMSE) and mean relative error (MRE) are used to evaluate the prediction performance of the two models. Table I shows the performance comparison of the two models on training dataset and test dataset.

TABLE I. PERFORMANCE COMPARISON OF THE PROPOSED MODEL AND THE SVR MODEL

Model	RMSE /Training	RMSE /Test	MRE /Training	MRE /Test
Proposed	5393	7671	0.19	0.22
SVR	9024	10216	0.20	0.28

From Figs. 2 - 5, it can be found that both the proposed model and the SVR model fit the training data well. And the fitting effects of the two models on the test dataset are inferior to the training data, which are consistent with the change curves of training loss and test loss. The predicted value of the proposed model is basically consistent with the true value while there is a big deviation between the predicted value and the real value of some data for the SVR model. As shown in Table I, both the RMSE and MRE indicators of the proposed model are smaller than those of the SVR model on training dataset and test dataset, especially the RMSE indicator. Therefore, the proposed

model achieves better prediction performance results, compared with the SVR model.

B. Prediction of House Price Trend

From Section II.C, three parameters, i.e. p , d and q , have to be determined first in order to use the ARIMA model. The ACF (autocorrelation function) plot and the PACF (partial autocorrelation function) plot are applied to identify the parameters p and q . From the plots, it is clear that the values of p and q should be 2 and 1 respectively. In addition, the thermodynamic chart is employed to validate the parameters. Both the AIC and BIC criteria validate the correctness of the parameter settings for p and q . From the first order difference diagram, it can be found that there is no obvious increasing or decreasing trend in the difference diagram. And there is certain regularity for the volatility of differential data. Therefore, the value of d is set to be 1. For space reason, the ACF plot, PACF plot and the first order difference diagram are not given in this paper.

Fig. 6 shows the trend of house price, where the blue curve denotes the real data of each week from February 2017 to May 2018 and the red line denotes the predicted trend of each week from December 25, 2017 to August 30, 2018. From Fig. 6, it can be found that the predicted house price presents a light decreasing trend. The future price within one or two months predicted by the ARIMA model is basically consistent with the real data. But with the increase of prediction date, the predicted future house price would become inconsistent with the real data because the ARIMA model is a short-time prediction model.

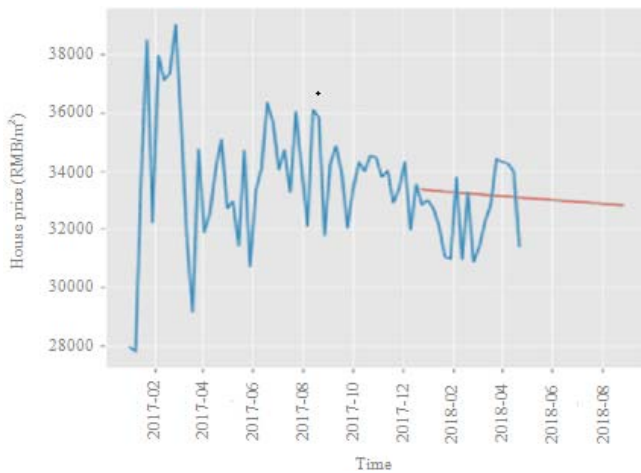


Fig. 6. The trend of house price.

Fig. 7 shows the quantile-quantile plot. It is obvious that the predicted data and the actual data coincide with the same distribution and the sample data is an approximately normal distribution.

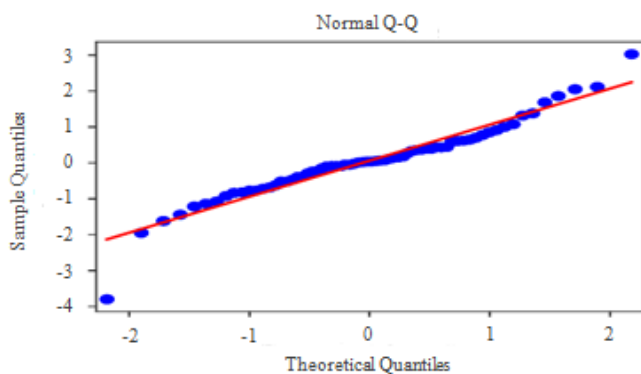


Fig. 7. The quantile-quantile plot.

Fig. 8 shows the standardized residual plot. It can be found that the residuals are distributed around the centerline randomly. Therefore, the model is believable according to the statistical theory.

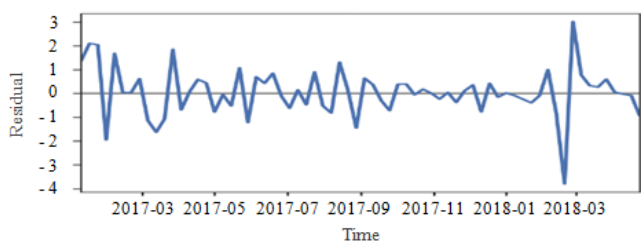


Fig. 8. The residual plot.

IV. CONCLUSION

The relationship between house price and influential factors is complex and nonlinear. And the traditional house price prediction approaches are lack of capacity for massive

data analysis. To address these concerns, a house price prediction approach based on deep learning and ARIMA model is proposed in this paper. The house price prediction model is implemented on the TensorFlow framework. And thirteen explanatory features are firstly selected to be the influential factors of house price. Then the raw housing data are crawled from Internet by using Scrappy, followed by the step of data preprocessing. Based on the extracted housing data, comparative experiments were conducted between the proposed approach and a SVR method. The experimental results show that individual house price predicted by the proposed approach is better than that of SVR method. And the predicted house price trend is basically consistent with the real data when generating short-run prediction.

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